ADITYA UNIVERSITY

Data Analysis Essentials

P.Srilatha

Assistant Professor

Department of IT & DS

Aditya University

Lab Practice

Practice:

Week 4: Indexing and Slicing of NumPy Array

a. Slicing 1-D NumPy arrays

b. Slicing 2-D NumPy arrays

c. Slicing 3-D NumPy arrays

d. Negative slicing of NumPy arrays

ADITYA UNIVERSITY

Indexing

Indexing is used to access individual elements in a NumPy array, while slicing is used to access a range of elements. Slicing helps in extracting subsets of arrays without modifying the original array.

Syntax

array[start : stop : step]

start – Starting index (inclusive)

stop – Ending index (exclusive)

step – Interval between elements (optional)

Slicing 1-D NumPy Arrays

a. Slicing 1-D NumPy Arrays

A 1-D array contains elements arranged in a single row.

Program

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50, 60, 70])
print(arr[1:5])
```

Output

```
[20 30 40 50]
```

Different Slicing Operations

program

```
import numpy as np
arr = np.array([10,20,30,40,50,60,70])
print("Original Array:", arr)
print(arr[:4])      # First four elements
print(arr[2:])      # From index 2 to end
print(arr[::2])     # Every second element
print(arr[1:6:2])   # Step size 2
```

Output

```
Original Array: [10 20 30 40 50 60 70]
[10 20 30 40]
[30 40 50 60 70]
[10 30 50 70]
[20 40 60]
```

Slicing 2D Arrays

b. Slicing 2-D NumPy Arrays

- A 2-D array consists of rows and columns.

Program

```
import numpy as np
arr = np.array([[1,2,3],
                [4,5,6],
                [7,8,9]])
print(arr)
```

```
[[1 2 3]
[4 5 6]
[7 8 9]]
```

Slicing 2D Arrays

Accessing Rows

print(arr[1])

[4 5 6]

Accessing Columns

print(arr[:,1])

[2 5 8]

Slicing Rows and Columns

print(arr[0:2, 1:3])

[[2 3]

[5 6]]

Slicing 2D Arrays

Examples

```
print(arr[1:, :2])
[[4 5]
[7 8]]
print(arr[:, ::2])
[[1 3]
[4 6]
[7 9]]
```

Slicing 3-D NumPy Arrays

c. Slicing 3-D NumPy Arrays

A 3-D array contains multiple 2-D arrays.

```
import numpy as np
arr = np.array([
    [[1,2,3],
     [4,5,6]],
    [[7,8,9],
     [10,11,12]]
])
print(arr)
```

```
[[[ 1 2 3]
[ 4 5 6]]
[[ 7 8 9]
[10 11 12]]]
```

Slicing 3-D NumPy Arrays

Shape of the array

```
print(arr.shape)
```

(2, 2, 3)

Accessing the First 2-D Array

```
print(arr[0])
```

[[1 2 3]

[4 5 6]]

Accessing an Element

```
print(arr[1,0,2])
```

9

ADITYA UNIVERSITY

Slicing

```
print(arr[:, :, 1:])
```

```
[[[ 2 3]
[ 5 6]]
[[ 8 9]
[11 12]]]
```

Negative Slicing of NumPy Arrays

d. Negative Slicing of NumPy Arrays

Negative indexing starts counting from the end of the array.

Index	Value
-1	Last element
-2	Second last
-3	Third last

Program

```
import numpy as np
arr = np.array([10,20,30,40,50,60,70])
print(arr[-1])
print(arr[-3])
```

Output

```
70
50
```

Negative Slicing of NumPy Arrays

Negative Slicing

```
print(arr[-5:-2])
```

[30 40 50]

Reverse an Array

```
print(arr[::-1])
```

[70 60 50 40 30 20 10]

Reverse Every Alternate Element

```
print(arr[::-2])
```

[70 50 30 10]

Negative Slicing in 2-D Arrays

Program

```
import numpy as np
arr = np.array([[1,2,3],
                [4,5,6],
                [7,8,9]])
print(arr[-2:, -2:])
```

Output

```
[[5 6]
[8 9]]
```

ADITYA UNIVERSITY

Thank You